

Program-9 : Organize Kubernetes Pods Using Labels and Namespaces

Description :

This program demonstrates how to organize Kubernetes Pods using labels and namespaces. Namespaces are used to separate environments such as development and production, and labels are used to categorize pods like frontend and backend.

Step-1 : Create Project Directory

```
mkdir program-9  
cd program-9
```

Step-2 : Create Namespace YAML File

```
nano namespaces.yaml
```

```
apiVersion: v1  
kind: Namespace  
metadata:  
  name: dev  
---  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: prod
```

Step-3 : Create Pods YAML File

```
nano pods.yaml
```

```
apiVersion: v1  
kind: Pod  
metadata:  
  name: frontend-pod  
  namespace: dev  
  labels:  
    app: frontend  
    env: dev  
spec:  
  containers:  
    - name: frontend  
      image: nginx  
      ports:  
        - containerPort: 80  
---  
apiVersion: v1  
kind: Pod
```

```
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Step-4 : Apply Namespace Configuration

kubectl apply -f namespaces.yaml

Step-5 : Apply Pods Configuration

kubectl apply -f pods.yaml

Step-6 : Verify Pods in Dev Namespace

kubectl get pods -n dev

Step-7 : Verify Pods in Prod Namespace

kubectl get pods -n prod

Step-8 : View Pods in All Namespaces

kubectl get pods --all-namespaces

Step-9 : Filter Pods Using Labels

kubectl get pods -n dev -l app=frontend

kubectl get pods -n prod -l app=backend

```
pavan-k@pavank1RV24MC072:~$ mkdir program-9
cd program-9
pavan-k@pavank1RV24MC072:~/program-9$ nano namespaces.yaml
pavan-k@pavank1RV24MC072:~/program-9$ nano pods.yaml
pavan-k@pavank1RV24MC072:~/program-9$ nano pods.yaml
pavan-k@pavank1RV24MC072:~/program-9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
pavan-k@pavank1RV24MC072:~/program-9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
pavan-k@pavank1RV24MC072:~/program-9$ kubectl get pods -n dev
kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           10s
frontend-pod   1/1     Running   0           10s
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    0/1     ContainerCreating 0           10s
frontend-pod   1/1     Running   0           10s
pavan-k@pavank1RV24MC072:~/program-9$ kubectl get pods --all-namespaces
NAMESPACE     NAME          READY   STATUS    RESTARTS   AGE
default       program-8     1/1     Running   0           9m23s
dev           backend-pod    1/1     Running   0           23s
dev           frontend-pod   1/1     Running   0           23s
kube-system   coredns-66bc5c9577-9s479 1/1     Running   3 (24m ago) 43m
kube-system   etcd-minikube 1/1     Running   3 (24m ago) 43m
kube-system   kube-apiserver-minikube 1/1     Running   3 (24m ago) 43m
kube-system   kube-controller-manager-minikube 1/1     Running   4 (24m ago) 43m
kube-system   kube-proxy-7n5h8 1/1     Running   3 (24m ago) 43m
kube-system   kube-scheduler-minikube 1/1     Running   3 (24m ago) 43m
kube-system   storage-provisioner 1/1     Running   5 (24m ago) 43m
prod          backend-pod    1/1     Running   0           23s
prod          frontend-pod   1/1     Running   0           23s
pavan-k@pavank1RV24MC072:~/program-9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0           32s
pavan-k@pavank1RV24MC072:~/program-9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           45s
pavan-k@pavank1RV24MC072:~/program-9$
```

Step-10 : Cleanup Resources

kubectl delete pods --all -n dev

kubectl delete pods --all -n prod

kubectl delete svc --all -n dev

kubectl delete svc --all -n prod

```
kubectl delete deployments --all -n dev  
kubectl delete deployments --all -n prod
```

Result :

Kubernetes Pods were successfully organized using namespaces and labels. Pods were isolated by environment and filtered efficiently using labels.